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Cross-scenario automatic sleep stage classification using transfer learning and single-channel EEG

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ABSTRACT

Sleep staging via manual inspection is time-consuming and inefficient, and as such, automatic sleep stage classification methods have been proposed and successfully implemented. However, these conventional supervised learning models are not suitable for cross-scenario sleep staging tasks due to the shift data distribution problem. This study aims to explore the use of transfer learning to transfer knowledge from a labeled source dataset to a new target domain where labels are difficult to obtain, so as to help solve the classification problem in a new domain in practical sleep staging scenarios. A novel end-to-end deep learning network that includes a feature extractor, a sleep stage classifier, and a domain adaptation network was proposed. The proposed domain adaptation network can accomplish distribution alignment between source and target domains, through the application of both joint distribution loss and stage transition loss. Cross-channel, cross-subject and channel, and cross-dataset experiments were designed to verify the feasibility of the proposed network using three publicly available datasets. The results indicate an average accuracy improvement ranging from 0.1% to 6.1% (average of 2.9%), a macro-averaging F1-score improvement ranging from 0.1% to 5.1% (average of 2.6%), and a Cohen's kappa coefficient improvement ranging from 0.016 to 0.083 (average of 0.045) after using transfer learning, when compared with a non-transfer model. Relative to similar existing transfer learning methods for sleep staging, the proposed method was implemented in an unsupervised manner and achieved success in cross-scenario sleep staging tasks, which is more suitable for practical applications in daily life.

1. Introduction

Sleep staging is an important method to ensure the effective diagnosis of sleep disorders and sleep quality assessment. The procedure for sleep staging includes the use of polysomnograph (PSG) technology to synchronously record the electroencephalogram (EEG), electrocardiogram (ECG), electromyogram (EMG), electrooculogram (EOG), etc., during sleep [1,2]. Experienced human experts can then complete the sleep staging process based on the above PSG records, together with staging standards via visual inspection. However, this is an extremely tedious process and is also prone to human error [3]. Therefore, automating sleep staging has become an important research topic with the purpose of supporting and improving diagnosis' efficiency.

More recently, automatic sleep stage classification based on single-channel EEG via supervised learning has been widely explored and successfully applied, yielding new machine learning methods and deep learning methods [4–16]. However, sleep staging classification tasks in the real world can be more challenging; for instance, it is generally time-consuming and expensive to have labeling done by human experts for many portable wearable EEG acquisition devices. Thus, one intuitive and feasible approach is to use publicly-available labeled records to achieve sleep staging. Such supervised learning methods are generally trained and tested on the same dataset. One crucial reason for this is that conventional supervised learning can learn better if the training and test data are drawn from the same feature space and the same distribution [17,18]. Therefore, the above supervised learning methods may suffer

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performance degradation when directly applied to the task in cross-scenario situations. When considering how to use the existing datasets (source) to subject the unlabeled dataset (target) to classification, the most common scenarios are as follows: (1) Source and target datasets have various EEG signal channels caused by the different positions of the electrodes on the scalp surface during data collection; and (2) source and target datasets are typically collected from diverse populations, devices, and software, so there are more significant discrepancies [19]. In summary, the main challenge in the above scenarios is that source and target datasets have the problem of shift data distribution; thus, the conventional supervised learning model trained on the existing dataset cannot be directly generalized to the target dataset for sleep staging.

Unsupervised domain adaption (UDA) [17], a well-known branch of transfer learning, has been regarded as a feasible method to settle the problem and was applied in this study. Transfer learning aims to learn from one dataset (called the source domain) and transfer the knowledge to a related but not identical dataset (called the target domain), achieving the distribution adaptation of the source domain and the target domain, whose potential has been extensively proven in abundant classification tasks [20–22]. In this study, it was attempted to apply transfer learning technology to solve the problem of cross-scenario automatic sleep staging, with the model being inspired by a popular transfer learning model called DANN [23]. However, DANN only aligns the global distribution between domains, and it may be insufficient to be applied to sleep staging because of the complexity and diversity in actual application scenarios, and thus matching the whole source and target without exploiting the discriminative structures may not work well in some tasks [24]. Previous studies have demonstrated that the simultaneous alignment of both the marginal and conditional distributions between domains [25,26], or subdomain adaptation [24,27] to align domain-specific distributions can both further contribute to the adaptation. Unlike them, in this study, a joint distribution alignment structure was designed to achieve the target, besides, a stage transition loss was also integrated into the model to further align the distribution among adjacent sleep stages. The performance was verified via three groups of experiments, and the results indicated that this can achieve an advantageous performance. The novelty and contributions of this study are the following:

(1) An end-to-end deep learning network that included a feature extractor, a sleep stage classifier and a domain adaptation network for cross-scenario automatic sleep staging. The model can transfer the knowledge learned from the source domain to the target domain mainly by the domain adaptation network, thereby assisting the sleep staging in a new unlabeled target domain.

(2) A joint distribution loss and a stage transition loss were implemented in the domain adaptation network of the model. The joint distribution loss can align the joint distribution of the category and the hidden feature of the current EEG epoch; the stage transition loss was inspired by the transition probability of adjacent sleep stages, which can effectively help to adapt the distribution of adjacent epochs between domains.

2. Related work

2.1. Supervised learning for sleep staging

In previous reported machine learning methods, experts have manually extracted the relevant features of sleep staging, according to domain knowledge. These include features extracted via wavelet transform [11], Fourier transform [28], empirical mode decomposition [9,29], the entropy method [10,30,31], etc. The manually extracted features were then selected as input to establish a machine learning classifier, such as Support Vector Machine [10,11], neural network [9], Random Forest classifier [14], ensemble learning model [15], etc.

Manual feature extraction can be challenging due to the non-stationarity and non-linearity of EEG, leading to the need of research focused on deep learning to model sleep staging. In particular, original signals were directly considered as input of neural networks, and stage labels were used as output for supervised learning. Models such as Convolutional Neural Network (CNN) [4,5,7,12,16], Recurrent Neural Network (RNN) [6], and a hybrid structure of CNN and RNN [8,13] have been reported in recent literature. Deep learning methods preclude the cumbersome process of manual feature extraction and can automatically learn potential representations, which has enabled even better performance compared to some traditional machine learning methods in sleep staging.

2.2. Transfer learning

Transfer learning has received widespread attention in recent years. The purpose of transfer learning is to use the knowledge learned from the source domain to assist with learning in a new target domain, so as to improve the performance of classification tasks in the target domain. Conventional supervised learning requires that both training and test datasets are drawn from the same feature space and have a highly similar marginal distribution. However, transfer learning has overcome this limitation, no longer requiring these two datasets to have similar marginal distributions, nor requiring that the feature space, label space, and conditional probability distribution to be identical, according to a study by Pan *et al.* [17]. Fine-tuning is a popular method in neural network-based transfer learning models [32–34], and its principal idea is that if both the source and target domain have labels, the neural network can be pre-trained on the source dataset. The specific lower layers are frozen, and the top layers are re-trained using the target dataset with a relatively small learning rate, after which a new mapping is established so the model is adaptable to the target task. Fine-tuning requires both source and target domains to have labels; however, in many practical application scenarios, it is easy to obtain labeled source domain datasets but difficult to obtain labeled target domain datasets. For example, PSG records collected in a hospital can be well labeled by human experts, but in a home environment, it is expensive to obtain sleep staging labels to situations where patients use portable EEG devices to perform sleep staging.

To solve this problem, UDA has become an important research branch in transfer learning. A maximum mean discrepancy (MMD) approach was proposed by Gretton *et al.* to measure the difference between two distributions [35], through the optimization of which distribution alignment of features can be achieved. Based on this, researchers further proposed several variant algorithms, for example, Tzeng et al. proposed a DDC method [36], where the MMD was introduced into a certain layer and then the MMD loss was combined with the classification loss to conduct a synchronized optimization. In this way, the neural network can not only learn the domain invariant representation, but also to complete the training of the classifier. However, DDC only completed the adaptation of one layer in the neural network, while ignoring the transferability of multiple layers. Because of this, Long et al. proposed a DAN method [37], where the representation of the original input was first extracted by deep network, and multiple layers and Multi-Kernel MMD were then applied to the process of adaptation. They further proposed a JAN [38], which is able to align the distribution of the feature representation and the label in a joint manner. Wang et al. proposed a STL algorithm [39], which was different from the traditional MMD only to measure the difference in the marginal distribution. In this work, an intra-class MMD was proposed, in order to calculate the distribution distance between each class, as well as to realize the knowledge transfer between domains at the class level. In contrast to MMD, statistic feature alignment is another widely used feature alignment approach. The CORAL method [40], for example, can learn a linear transformation by aligning the second-order statistics of source and target distributions to minimize domain shift. In addition, the idea of generative adversarial nets [41] has also been applied to transfer learning tasks, such as DANN [23] and ADDA [42]. DANN consists of a

shared feature extractor, a label predictor and a domain classifier. After the collaborative optimization of the classification loss and domain discrimination loss, the shared feature extractor can make it difficult for the generated feature representation to be distinguished by the domain classifier, that is, the distribution alignment of the source and target domain is realized, and then the target domain input can be classified by the label predictor. Different from DANN, ADDA provides separate feature extractors for the source and target domains, adopting the parameters pre-trained on the source domain to initialize the target domain feature extractor, which is more flexible than DANN because it allows the feature extractor to extract domain-specific representation. There are numerous other typical UDA methods, for example, DRCN proposed by Ghifary et al. [43], where a shared CNN is applied for encoding the original input, a branch is then stacked for supervised classification based on the labeled data from the source domain, while another branch is designed as a decoder to reconstruct the original input.

2.3. Sleep staging using transfer learning

The process of applying transfer learning to sleep staging has also attracted research attention. Vilamala *et al.* used multitaper spectral analysis to transform original EEG signals to visual images, applying fine-trained deep CNN on image classification tasks to solve sleep staging tasks [44]. This method, which can be considered as a "fine-tuning" type of transfer learning, yielded better performance than the network trained from scratch. Andreotti *et al.* performed subject-specific "fine-tuning" on their own collected dataset [45], they first pre-training the model on two publicly released datasets and then transferring the model to their own dataset, showing an improved performance compared to not using transfer learning. However, the performance varies with the applied basic neural network, so further validation is needed. In addition, to build a model to predict the results for the second night, the original records of the first night still must be manually labeled. Zhang *et al.* proposed a transfer learning algorithm based on deep auto-encoders [18], which can learn the potential representation of the original signal in an unsupervised manner, while simultaneously measuring the divergence in the hidden layer between two probability distributions by the Kullback-Leibler divergence. The model achieved a higher performance than both baseline and two traditional transfer learning methods, which demonstrated that transfer learning methods in this task were superior to non-transfer methods and the potential of the combination of transfer learning and deep learning. However, the own-collected dataset was only from 10 subjects, and only 3 sleep stages were distinguished; thus, the results still must be further validated by more experiments. Jadhav *et al.* used the continuous wavelet transform technique to represent the EEG signal as a time–frequency image [46]; a deep CNN model pre-trained on an image dataset that was used to fine-tune and conduct sleep staging classification.

The above research made full use of data from other domains to build sleep-scoring models. However, such methods based on "fine-tuning" still require labeled target domain samples so that the model can adapt the knowledge learned from the source dataset to the target domain. In this study, the feasibility of establishing a UDA model to accomplish the knowledge transfer in an unsupervised manner was explored, which is more suitable for practical application scenarios.

3. Problem formulation

EEG signals with channel C and length L in the source and the target domains are denoted as $\boldsymbol{x}^s \in \mathbb{R}^{C\times L}$ and $\boldsymbol{x}^t \in \mathbb{R}^{C\times L}$, respectively. $P(\boldsymbol{x}^s)$ and $P(\boldsymbol{x}^t)$ are the marginal distributions. The labeled source dataset is $D^s = \{(x_1^s, y_1^s), \ldots, (x_{n^s}^s, y_{n^s}^s)\}$, and the unlabeled target dataset is $D^t = \{x_1^t, \ldots, x_{n^t}^t\}$, where n^s and n^t are the number of samples, and labels in D^t cannot be accessed during training. Both domains have the same feature space and label space, but $P(\boldsymbol{x}^s) \neq P(\boldsymbol{x}^t)$. The goal of our proposed algorithm is to obtain the labels y^t for the unlabeled target domain, using the labeled source domain data.

4. Dataset

Three publicly available datasets, Sleep-EDF Expanded, Sleep Heart Health Study 1 (SHHS1) and PhysioNet Challenge 2018, were used to evaluate the proposed method. The final frequency of each 30-second sleep epoch is summarized in Table 1.

Sleep-EDF-20 dataset (EDF-set): The 2013 version of the Sleep-EDF Expanded database includes a total of 20 subjects (10 males and 10 females, between 25 and 34 years old), each individual has a record of two nights, and 39 separate records can be used due to subject 13 having only one record [47,48]. Each record contains the Fpz-Cz and Pz-Oz EEG channels, 1 EOG (horizontal), 1 EMG, and 1 respiration signal; the sampling rate of the EEG signal is 100 Hz. The recordings are split into short periods of 30 s called epochs, which are then categorized into eight classes, including Wake, REM, N1, N2, N3, N4, MOVEMENT, and UNKNOWN according to the R&K standard [49]. This study was exclusively focused on single-channel Fpz-Cz or Pz-Oz EEG signals and merged N3 and N4 to a single stage according to the AASM standard [50] to optimally compare the performance with similar research. MOVEMENT and UNKNOWN classes were excluded because of their irrelevance to the task. Thirty minutes before sleep status and thirty minutes after sleep status were excluded according to previous literature [51].

SHHS1-100 dataset (SH-set): The SHHS dataset was collected from a multi-center study to investigate the effect of sleep-disordered breathing on cardiovascular effects and other consequences. It includes SHHS visit 1 (SHHS1) and visit 2 (SHHS2), the length of each epoch is also 30 s, and the R&K criteria were adopted to manually label the data. The sampling rate of the EEG signal is 125 Hz [52,53]. In this study, the only focus was on the SHHS1 dataset, which contains a total of 5,793 subjects, but only the records of the first 100 individuals were included for modeling, to reduce the computational load. In addition, it was referred to the general practice of other studies and merged the N3 and N4 stages, and only considered the single-channel C3-A2 and the C4-A1 EEG signal channel.

PhysioNet-100 dataset (PN-set): This dataset was provided by the PhysioNet Computing in Cardiology (CinC) Challenge 2018 [48,54], having been collected at Massachusetts General Hospital's sleep laboratory, and containing PSG recordings of 1,985 subjects. The dataset had sleep stages labels assigned by certified human experts using the AASM guideline, including W, N1, N2, N3 and REM, and the signals were recorded at a sampling rate of 200 Hz. Ultimately, 100 subjects were randomly collected from the publicly available training set (994 subjects) due to computational resources limitations, the selecetd signal channel for experiment was C3-M2.

5. Methods

The overall framework of the proposed model is illustrated in Fig. 1, which include feature extraction, classifier training, and domain adaptation in a joint manner.

Table 1
The frequency of each 30-second sleep stage epoch on Sleep-EDF-20 (EDF-set), SHHS1-100 (SH-set) and PhysioNet-100 (PN-set) datasets.

Dataset	Wake	REM	N1	N2	N3 + N4 (SWS)
Sleep-EDF-20 (EDF-set)	9,155	7,717	2,804	17,799	5,703
SHHS1-100 (SH-set)	24,240	11,703	3,245	38,430	12,933
PhysioNet-100 (PN-set)	13,726	11,296	9,637	36,531	11,064

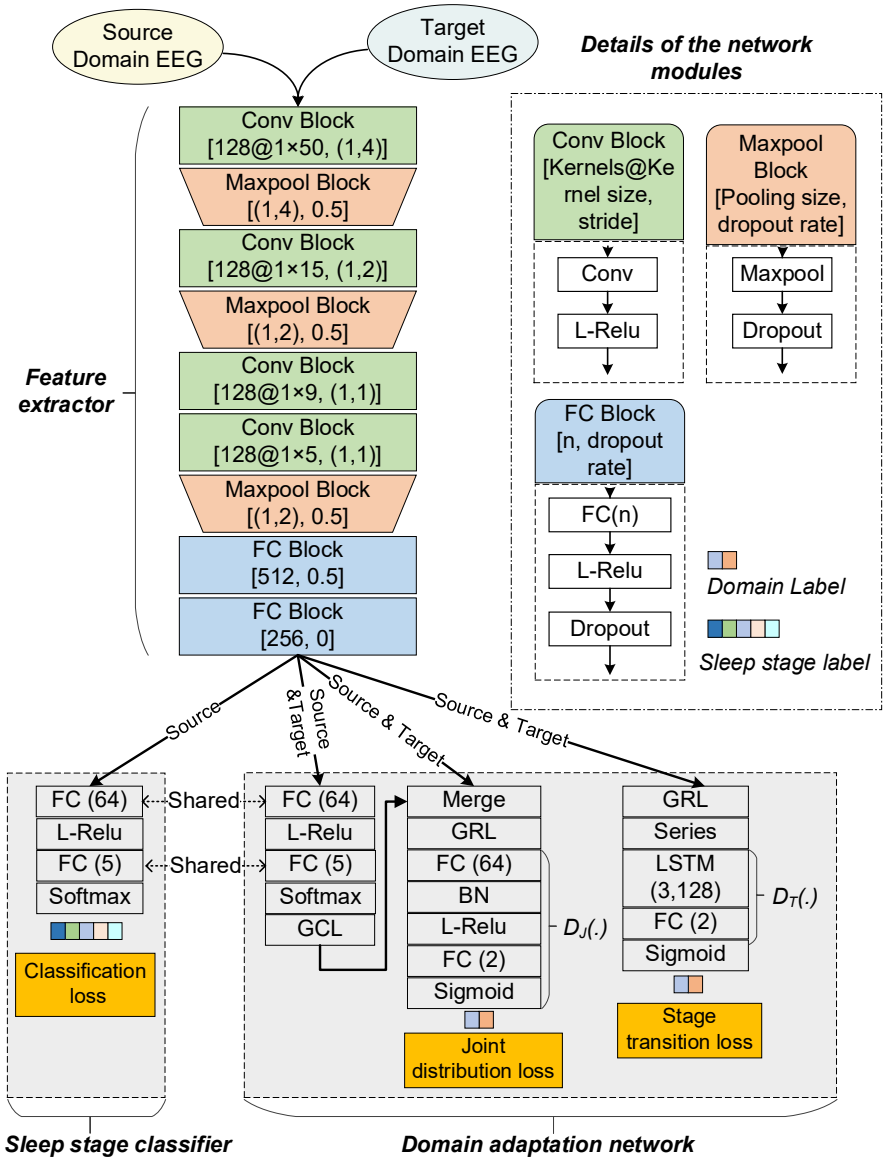


Fig. 1. The overall framework of the proposed model. This model includes a feature extractor, a sleep stage classifier, and a domain adaptation network. The overall loss function is composed of a classification loss, a joint distribution loss, and a stage transition loss. Parameters of the network are optimized in an end-to-end manner with a stochastic gradient descent (SGD) algorithm. Conv: Convolutional layer, L-Relu: Leaky Relu, FC: fully connected, BN: Batch normalization, GRL: gradient reversal layer, GCL: gradient cut-off layer, LSTM: long short-term memory.

5.1. Feature extractor

The feature extractor is defined as $E(\cdot)$, and the process of the feature extraction is $\boldsymbol{f} = E(\boldsymbol{x}; \theta_E)$, where $\boldsymbol{x} = \boldsymbol{x}^s \cup \boldsymbol{x}^t$ and θ_E are the parameters of the feature extractor and should be learned during model training. Many existing studies first transformed the EEG signal into a power spectrum using Fourier transform [12,55], but the proposed feature extractor captures the latent representation directly from the original EEG waveform; in particular, the EEG can be regarded as 2-D input with two dimensions. These are called as "lead" and "sample," and for a single-channel EEG, the "lead" dimension can be regarded as 1 and can be easily implemented using a current popular framework such as *pytorch* [56].

The feature extractor is composed of three types of blocks on the basis of a convolutional neural network, whose definitions are presented in Fig. 1. EEG signals from different datasets are uniformly resampled to 100 Hz so that the feature extractors can be shared. It is worth noting that previous practice has demonstrated that a large convolution kernel can have a larger receptive field and capture frequency domain information in the EEG signal [8]. Therefore, a large convolution kernel with a size of 1×50 is used at the first layer, and then gradually reduced. In addition, a stride and pooling size of (1, 4) was applied, which can contribute to reducing the size of the neural network and speed up the model's convergence. A dropout strategy with a probability of 0.5 is applied after each Maxpool layer, which can help to prevent over-fitting and improve the generalization performance of the model [57].

5.2. Sleep stage classifier

The feature vector $\boldsymbol{f}^s$ from the source domain is further passed to the classifier $C(\cdot)$ and yields the category prediction $\boldsymbol{y}^s = C(\boldsymbol{f}^s; \theta_C)$, where θ_C is the parameter of the classifier. This work was exclusively focused on five-class sleep staging, so the label of one epoch is $y_i \in \{Wake, REM, N1, N2, SWS\}$. $C(\cdot)$ is composed of 1 hidden fully-connected (FC) layer with 64 neurons and an output FC layer. The loss of the classifier is as follows:

$$\mathscr{L}_{\mathrm{cls}} = \frac{1}{\mathrm{n^s}} \sum_{\mathrm{i=1}}^{\mathrm{n^s}} \mathrm{L_c}\left(\mathrm{C}\left(\mathrm{E}\left(\mathrm{x_i^s}\right)\right), \mathrm{y_i^s}\right) \tag{1}$$

where L_c is the cross-entropy loss function.

5.3. Domain adaptation network

Another branch of the network is composed of two components, which were designed to accomplish the UDA.

5.3.1. Joint distribution loss

Inspired by DANN [23], the hidden layer output of the source and target data on the feature extractor are used for adversarial learning. Adversarial learning achieves the distribution alignment of the source and target domain by constructing a suitable domain discriminator, so that the target domain can use the classifier that was indiscriminately trained on the source domain. However, the loss function proposed in DANN only solved the problem of the marginal distribution alignment between source and target domains, while neglecting the difference in the distribution within the class. Other studies introduced the conditional distribution loss to measure the intra-class distribution divergence [25,26]. They adopted the pseudo-label method to indicate the target domain data label and then separately calculating the domain loss within each category to realize class-wise alignment.

In contrast to these studies, a joint distribution loss is proposed in this study to solve the problem. That is, the output of the feature extractor is first obtained, and then a network that fully shares parameters with the source sleep stage classifier is constructed to predict the probability of the target domain data. However, it should be noted that if we directly optimize the network, the gradient caused by the domain adaptation network will also be synchronously updated to the source classifier due to the existence of shared parameters. Since the target dataset has no labels during training, the parameters of the classifier should be only updated by the labeled source dataset (i.e., using $\mathscr{L}_{\mathrm{cls}}$). To solve this problem, this study proposes an innovative design by using a gradient cut-off layer (GCL). This is an identical mapping in the feed forward process, but it cuts off the gradient (set it to 0), which can effectively terminate the gradient from the domain adaptation network to the source classifier:

$$x = GCL(x) \tag{2}$$

$$\frac{dGCL(x)}{dx} = 0 \tag{3}$$

The output of the network with shared parameters is as follows:

$$\widehat{\boldsymbol{y}} = GCL(C(E(\boldsymbol{x}))) \tag{4}$$

A discriminator $D_J(\cdot;\theta_{DJ})$ is then built, which is used to distinguish the input from the source or target domain; this network is stacked by 2 FC layers, with a batch normalization layer (BN) [58] and leaky rectified linear activation unit (LeakyReLU). If we define $d_i = 1$ if$x_i \in D^s$, and $d_i = 0$ if$x_i \in D^t$, the loss is the following:

$$\mathscr{L}_{\mathrm{joint}} = \frac{1}{\mathrm{n^s + n^t}} \sum_{\mathrm{x_i} \in (\mathrm{D^s \cup D^t})} \mathrm{L_d}\left(\mathrm{D_J}\left(\mathscr{M}\left(\mathrm{E}(\mathrm{x_i}), \widehat{\mathrm{y}}_{\mathrm{i}}\right)\right), \mathrm{d_i}\right) \tag{5}$$

where $\mathscr{M}(\cdot,\cdot)$ is used to merge (or concatenate) the features and label prediction probability, $\widehat{y}_i$ is the prediction probability for the ith sample, and L_d is the binary cross-entropy loss function.

5.3.1.1. Stage transition loss. The probability of sleep transition is an inherent feature of sleep staging. Using the transition probability of adjacent sleep epochs is an effective way to improve the performance of the current epoch's decision-making [51,59].

From the perspective of state transition, the probability of the current state of sleep transitioning to the next state is different. Fig. 2 illustrates the transition probability matrix counted on two datasets. For example, on the EDF-set, if the current state is the stage called Wake, the most possible state of the next stage is also Wake, with a probability of 0.942, while that of SWS is close to zero. Therefore, the ith feature vector f_i output by the feature extractor should not only account for the joint distribution of the category and the hidden feature of the current EEG epoch, but should also account for the joint distribution of adjacent sleep stages, so a sequence with length H as $\mathscr{S} = \left[\mathrm{f_{i-H}}, \ldots, \mathrm{f_i}\right]$ is first built, and then a discriminator $D_T(\cdot;\theta_{DT})$ based on the LSTM network [60] can be constructed. This will identify the similarity of the sequence to the source domain and target domain, thereby achieving the consistency of the distribution of the transition probability:

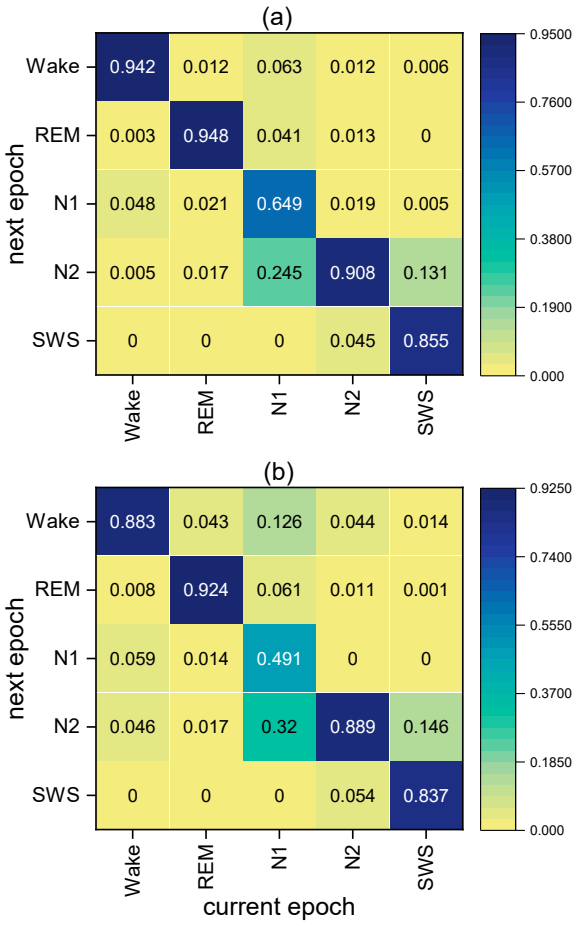


Fig. 2. Sleep stage transition probability matrix on: (a) Sleep-EDF-20 dataset (EDF-set); (b) SHHS1-100 dataset (SH-set).

$$\mathscr{L}_{\mathrm{trans}} = \frac{1}{\mathrm{n^s + n^t}} \sum_{\mathrm{x_i} \in (\mathrm{D^s \cup D^t})} \mathrm{L_d}\left(\mathrm{D_T}(\mathscr{S}_{\mathrm{i}}), \mathrm{d_i}\right) \tag{6}$$

In this study, H is set to 3, and the number of neurons in the hidden layer of the LSTM is 128.

5.3.1.2. Overall formulation and optimization. According to the aforementioned modules, the total loss is as follows:

$$\mathscr{L}_{\mathrm{total}}(\theta_{\mathrm{E}}, \theta_{\mathrm{C}}, \theta_{\mathrm{DJ}}, \theta_{\mathrm{DT}}) = \mathscr{L}_{\mathrm{cls}} - \gamma \cdot \mathscr{L}_{\mathrm{joint}} - \delta \cdot \mathscr{L}_{\mathrm{trans}} \tag{7}$$

where γ and δ are balance parameters for the total loss, and the goal of optimization is a mini-max process:

$$\widehat{\theta}_{DJ}, \widehat{\theta}_{DT} = arg \max_{\theta_{DJ}, \theta_{DT}} \mathscr{L}_{\mathrm{total}}(\widehat{\theta}_{\mathrm{E}}, \widehat{\theta}_{\mathrm{C}}, \theta_{\mathrm{DJ}}, \theta_{\mathrm{DT}}) \tag{8}$$

$$\widehat{\theta}_E, \widehat{\theta}_C = \underset{\theta_E,\theta_C}{argmin}\mathscr{L}_{\text{total}}(\theta_E, \theta_C, \widehat{\theta}_{DJ}, \widehat{\theta}_{DT}) \tag{9}$$

The $D_J(\cdot)$ and $D_T(\cdot)$ aim to distinguish data from opposite domains in (8); thus, they attempt to minimize $\mathscr{L}_{\text{joint}}$ and $\mathscr{L}_{\text{trans}}$, respectively, that is, to maximize $\mathscr{L}_{\text{total}}$. However, in (9), $E(\cdot)$ aims to obfuscate $D_J(\cdot)$ and $D_T(\cdot)$; thus, it aims to maximize $\mathscr{L}_{\text{joint}}$ and $\mathscr{L}_{\text{trans}}$, that is, to minimize $\mathscr{L}_{\text{total}}$. $C(\cdot)$ aims to accurately predict the sleep stages, so it aims to minimize $\mathscr{L}_{\text{cls}}$.

The above process can alternately perform to complete the optimization. However, to realize the end-to-end training and embed the loss in the network, the gradient reversal layer (GRL) design is referred in DANN. This has an identical mapping in regards to the feed-forward process:

$$x = R^{\lambda}(x) \tag{10}$$

The GRL layer can conduct the negative control of the gradient when the gradient is propagated backwards, and the degree can be adjusted by a coefficient λ:

$$\frac{dR^{\lambda}(x)}{dx} = -\lambda \tag{11}$$

The λ in GRL is dynamically adjusted from 0 to 1:

$$\lambda = \frac{2}{1 + e^{-\varepsilon * iter/iters}} - 1 \tag{12}$$

where ε is a coefficient to control λ, and *iter* and *iters* are the current training iteration and the total number of iterations, respectively.

The loss functions can be rewritten as the follows:

$$\overline{\mathscr{L}_{joint}} = \frac{1}{n^s + n^t} \sum_{x_i \in (D^s \cup D^t)} L_d\left(D_J\left(R_J^{\alpha}(\mathscr{M}(E(x_i), \widehat{y}_i))\right), d_i\right) \tag{13}$$

$$\overline{\mathscr{L}_{trans}} = \frac{1}{n^s + n^t} \sum_{x_i \in (D^s \cup D^t)} L_d\left(D_T\left(R_T^{\beta}(\mathscr{S}_i)\right), d_i\right) \tag{14}$$

$$\overline{\mathscr{L}_{\text{total}}(\theta_E, \theta_C, \theta_{DJ}, \theta_{DT})} = \mathscr{L}_{\text{cls}} + \overline{\mathscr{L}_{\text{joint}}} + \overline{\mathscr{L}_{\text{trans}}} \tag{15}$$

where R_J^{α} and R_T^{β} are GRL for $D_J(\cdot)$ and $D_T(\cdot)$, and thus the optimization target can be regarded as minimizing the $\overline{\mathscr{L}_{\text{total}}(\theta_E, \theta_C, \theta_{DJ}, \theta_{DT})}$. This network can synchronously conduct UDA and collaboratively accomplish the training of the classifier for the sleep staging task. Moreover, a standard stochastic gradient descent (SGD) algorithm can be used in the optimization process.

5.3.1.3. Prediction of the target domain label. Once the training of the model is completed, the learned domain invariant feature extractor $E(\cdot)$ can output indistinguishable hidden features, and thus the sleep stage classifier on a source dataset can be directly used for the classification of the target domain task.

6. Experiments

6.1. Experimental design

A baseline experiment was first defined to compare performances:

- **Baseline experiment:** Training and test data are derived from the same dataset and EEG channel, but the subjects are different. This is the situation most frequently used in traditional supervised learning methods. The performance is verified via 10-fold cross-validation.

Three groups of experiments are then designed to verify the performance of the proposed model. They are the most common scenarios in practical applications, as shown in Fig. 3(b).

- **Cross-channel experiment:** Training and testing data are derived from the same dataset and subject, but channels are different; one channel is used for the source domain data, and another channel from the same subject is used for the target domain data. This scenario primarily occurs when considering the different distribution of electrodes on the scalp surface. 30 records are randomly extracted with replacement from the dataset in each experiment, and the average results of 10 experiments are used to evaluate the model performance.
- **Cross-subject and channel experiment:** All the subjects in the same dataset are divided into 10-folds: 9 folds in one EEG channel are the source domain, leaving one fold with another channel as the target domain. If another channel is the same as the channel on the source domain, then the situation is the baseline we mentioned above.
- **Cross-dataset experiment:** Training and testing data are derived from different datasets, which means the subject and channel are also various, and this is the most challenging scenario. Different datasets may have differences between individuals, equipment, and EEG electrodes simultaneously. Similar to the above cross-channel experiment, 30 records with one EEG channel are randomly extracted with replacement from the dataset in each experiment, and the average results of 10 experiments are used to evaluate model performance.

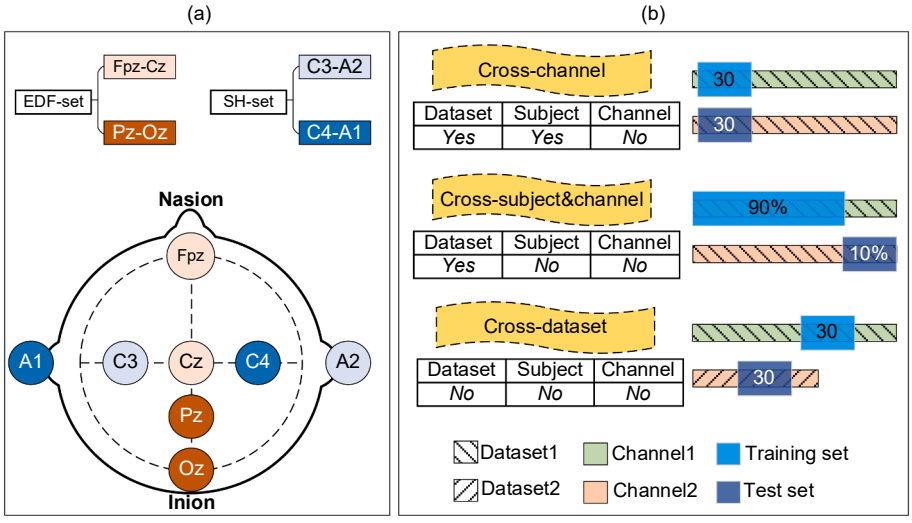


Fig. 3. Experimental design. (a) Electrode distribution on the scalp surface of the EEG signal from the Sleep-EDF-20 (EDF-set) and SHHS1-100 (SH-set) dataset. (b) Three groups of experiments with a total of 16 scenarios to validate the proposed method. Cross-channel experiment (Top): Training and test data are derived from the same dataset and subject, but the channels are different; Cross-subject and channel experiment (Middle): Training and test data are derived from the same dataset, but the subject and channel are different; Cross-dataset experiment (Bottom): Training and test data are derived from different datasets, and the subject and channel are also various.

It should be noted that due to the diversity of possible combinations, we mainly carried out the experiments on two datasets, including EDF-set and SH-set, the electrode distribution on the scalp surface of the EEG signal from these two datasets is illustrated in Fig. 3(a). Furthermore, we added a group of additional experiments based on the PN-set to further verify the model performance in cross-dataset scenario. Two paradigms were defined to compare the performance of the proposed method. The first paradigm involves the training of the model on the source dataset, and then directly evaluating the trained model on the test dataset (called no transfer); another paradigm uses our proposed transfer model to perform the UDA and then to evaluate the performance on the test dataset.

6.2. Model training

The model was trained in an end-to-end manner, using a standard SGD algorithm. The batch size was set to 512 (256 source data and 256 target data). The initial learning rate was 0.0001, and a dynamic learning rate scheduler (called ReduceLROnPlateau) was applied, in which the patient was set to 5, and the factor was 0.8. The ε in both the R_J^{α} and R_T^{β} layer were set to 4 in this study, and the total iterations were 50.

6.3. Performance evaluation

A macro-averaging F1-score (*MF*1), accuracy (*Acc*), and Cohen's kappa coefficient (k) were proposed for evaluation. They are defined as follows.

$$MF1 = \frac{\sum_{i=1}^{C} F1_i}{C}, \quad i = 1, \ldots, C \tag{16}$$

$$Acc = \frac{\sum_{i=1}^{C} TP_i}{N}, \quad i = 1, \ldots, C \tag{17}$$

$$k = \frac{p_o - p_e}{1 - p_e} \tag{18}$$

TP_i is the true positive for class$i$, and $F1_i$ is the F1-score for class$i$. p_o is the observed agreement, p_e is the hypothetical probability of chance agreement, C is the number of total sleep classes, and N is the number of total sleep stages. The 10-fold cross-validation was adopted to evaluate the experimental results, and the Wilcoxon signed-rank test was selected to compare the difference, where a p-value < 0.05 was considered statistically significant.

It should be emphasized that the division of training and testing sets in this study was strictly based on the individual rather than the epoch [51]. It has been proven that the division according to epoch instead of individual can yield a significantly higher performance [8,61]. However, this scheme is not practical in the real world, and it is believed that the essence of such performance improvement is due to over-fitting; for example, if one training sample and one test sample are from two consecutive epochs, then they have approximate EEG patterns, and thus a high performance can be achieved in the testing phase, but the learned hidden feature representation is difficult to generalize to a real independently divided test dataset.

7. Results and discussion

7.1. Baseline experiment results

The baseline results using 10-fold cross-validation on two datasets are presented in Table 2. On the EDF-set, the average performance obtained from the Fpz-Cz channel is higher than that of the Pz-Oz channel, and this finding is also in accordance with other related research [8,61]. The average k obtained from the Fpz-Cz channel is 0.764, which is an approximative performance compared to related works that used the same EEG channel [8,12,62]. On the SH-set, a maximum k of 0.688 was obtained from the C3-A2 channel, and the total performance results are lower than those on the EDF-set, indicating the diversity of baseline results on different datasets.

Table 2
Baseline experiment results on EDF-set and SH-set.

Dataset	Channel	*Acc*(%)	*MF*1(%)	k
EDF-set	Fpz-Cz	83.9	77.3	0.764
	Pz-Oz	82.0	70.7	0.739
SH-set	C4-A1	77.1	58.9	0.657
	C3-A2	79.0	61.5	0.688

7.2. Cross-channel experiment results

The performance results from the cross-channel experiment are indicated in Table 3. Three performance indicators improved after using transfer learning for all four scenarios, suggesting that even for the same individual, there are still differences in EEG signal channels caused by various electrode positions, and transfer learning is an effective approach to reduce discrepancies. As far as k is concerned, the optimal improvement in the EDF-set was Fpz-Cz → Pz-Oz, which increased by 0.062. In the SH-set, the best improvement was C3-A2 → C4-A1, which represented an increase of 0.055.

Compared with the baseline, the performance results of the two scenarios in the EDF-set dropped both before and after transfer, while performance dropped before transfer, but slightly increased after using transfer learning on the SH-set.

7.3. Cross-subject and channel experiment results

As presented in Table 4, the three performance indicators were enhanced after using transfer learning for all scenarios. As for k, the best improvement in the EDF-set was Pz-Oz → Fpz-Cz, which increased by 0.053. The best increase in the SH-set was C3-A2 → C4-A1, which increased by 0.057.

In addition, compared to the cross-channel scenario, the k before and after the transfer learning is lower, and it is mainly because in the cross-subject and channel scenario, there are not only differences in signal channels, but also in individuals. Moreover, compared with the cross-channel experiment, no matter which scenario, the EDF-set Pz-Oz → Fpz-Cz and SH-set C3-A2 → C4-A1 yielded the best performance after using transfer learning on their respective datasets.

7.4. Cross-dataset experiment results

In the following eight scenarios shown in Table 5, different levels of performance improvement were achieved through transfer learning, but

Table 3
Cross-channel experiment results in a total of four scenarios.

Dataset	Source channel → target channel	No transfer			Proposed transfer method		
		Acc (%)	*MF*1 (%)	k	*Acc* (%)	*MF*1 (%)	k
EDF-set	Fpz-Cz → Pz-Oz	72.8	64.4	0.635	77.8*	67.6*	0.697*
	Pz-Oz → Fpz-Cz	76.6	66.3	0.679	79.1*	68.1*	0.713*
SH-set	C4-A1 → C3-A2	74.9	60.1	0.636	77.1*	63.9*	0.674*
	C3-A2 → C4-A1	76.8	61.0	0.665	80.1*	64.5*	0.720*

* represents statistical significant ($p < 0.05$).

Table 4
Cross-subject and channel experiment results in a total of four scenarios.

Dataset	Source channel → target channel	No transfer			Proposed transfer method		
		Acc(%)	*MF*1(%)	*k*	*Acc*(%)	*MF*1(%)	*k*
EDF-set	Fpz-Cz → Pz-Oz	73.0	63.7	0.621	74.4*	64.3	0.652*
	Pz-Oz → Fpz-Cz	74.0	63.1	0.638	77.8*	65.0*	0.691*
SH-set	C4-A1 → C3-A2	72.4	55.9	0.611	74.9*	59.9*	0.641*
	C3-A2 → C4-A1	72.2	55.9	0.606	76.6*	60.5*	0.663*

* represents statistical significant ($p < 0.05$).

Table 5
Cross-dataset experiment result in a total of eight scenarios.

Source channel → target channel	No transfer			Proposed transfer method		
	Acc(%)	*MF*1(%)	*k*	*Acc*(%)	*MF*1(%)	*k*
EDF-set Fpz-Cz → SH-set C4-A1	66.8	55.5	0.525	68.7*	56.8*	0.579*
EDF-set Fpz-Cz → SH-set C3-A2	64.0	53.9	0.514	70.1*	58.6*	0.589*
EDF-set Pz-Oz → SH-set C4-A1	66.8	54.6	0.541	72.9*	59.7*	0.624*
EDF-set Pz-Oz → SH-set C3-A2	65.3	53.4	0.521	68.5*	55.3*	0.568*
SH-set C4-A1 → EDF-set Fpz-Cz	67.9	56.5	0.568	70.1*	56.6	0.594*
SH-set C4-A1 → EDF-set Pz-Oz	71.0	57.7	0.594	72.4	58.0	0.616*
SH-set C3-A2 → EDF-set Fpz-Cz	71.4	58.8	0.607	74.2*	60.1*	0.640*
SH-set C3-A2 → EDF-set Pz-Oz	72.1	57.9	0.609	72.2	58.6	0.625*

* represents statistical significant ($p < 0.05$).

the overall performance before and after the transfer learning are both lower than that of the cross-subject and channel scenario due to simultaneous differences in datasets, individuals, and channels. In addition, when the EDF-set was used as the source dataset, the EDF-set Pz-Oz → SH-set C4-A1 achieved the best performance and the greatest *k* performance improvement (0.083). And when the SH-set was used as the source dataset, the SH-set C3-A2 → EDF-set Fpz-Cz achieved the best performance and the greatest *k* improvement (0.033).

An additional cross-dataset experiment was conducted on the PN-set, the C3-M2 channel was applied as the source domain, and four channels from SH-set and EDF-set are used as the target domain. The results are shown in Table 6, it's found the proposed transfer learning method can still improve the performance in cross-dataset scenario.

7.5. Effectiveness of the proposed transfer model

From the above cross-channel, cross-subject and channel, and cross-dataset experiments, an *Acc* improvement ranging from 0.1 % to 6.1 % (average: 2.9 %), an *MF*1 improvement ranging from 0.1 % to 5.1 % (average: 2.6 %), and a *k* improvement ranging from 0.016 to 0.083 (average: 0.045) were achieved after the transfer learning was compared with the non-transfer model. On the one hand, this illustrates that our proposed transfer method has a positive effect on improving performance. On the other hand, it also illustrates that the non-transfer network still has learned knowledge with efficient generalization, so the improvement after transfer is limited. According to sleep staging rules such as the AASM principle [50], the EEG frequency is a key consideration for sleep staging, suggesting that different sleep datasets still have general character; thus, as long as the neural network is properly designed, the non-transfer model can also achieve adequate generalization on the target domain.

Table 6
Additional Cross-dataset experiment.

Source channel → target channel	No transfer			Proposed transfer method		
	Acc(%)	*MF*1(%)	*k*	*Acc*(%)	*MF*1(%)	*k*
PN-set C3-M2 → SH-set C4-A1	68.2	56.8	0.551	71.6*	59.5*	0.601*
PN-set C3-M2 → SH-set C3-A2	67.0	55.3	0.543	69.8*	58.7*	0.593*
PN-set C3-M2 → EDF-set Fpz-Cz	74.9	59.9	0.615	77.1*	62.9*	0.662*
PN-set C3-M2 → EDF-set Pz-Oz	74.0	58.3	0.608	75.5*	61.5*	0.643*

* represents statistical significant ($p < 0.05$).

There are numerous literatures that have explored the generalizability of models in different cohorts. Cesari et al. conducted experiments to evaluate interrater reliability for sleep stage scoring between human scorers from 2 European sleep centers and the automatic Stanford-STAGES algorithm in a cohort [63]. The results indicated the Stanford-STAGES algorithm had a good ability to generalize to new cohorts. Olesen et al. proposed a sleep staging algorithm on basis of deep learning [64], the prime modules included a residual network and a bidirectional gated recurrent unit, and the method was verified under different conditions. Different cohorts generalize very differently both in intra- and inter-cohort settings was found, and more data sources was useful for performance enhancement. Stephansen et al. proposed a convolutional and recurrent neural network to derive hypnodensity features for the diagnosis of Type-1 Narcolepsy [65], the models were evaluated on different cohorts, and the most successful model was found to be the one consisting of an ensemble of different models. In another study performed by Alvarez-Estevez et al., a novel flexible design combining different layers of CNN and LSTM was proposed [66]. This design added the capacity for learning of transition rules among sleep stages than the one only including CNN blocks. Besides, the proposed ensemble-based approach can enable flexible and scalable design, and further increase generalization capabilities. In summary, deep learning is a common strategy to improve generalization capabilities, one explanation is that deep learning can learn information that is difficult to be discovered by traditional artificial feature extraction. The backbone of our proposed neural network is similar to the aforementioned methods, which adopts the hybrid structure of CNN and LSTM. However, their models are only trained on the training set and does not perceive any information from the test set, so it's hard to further enhance the performance because the inherent differences in the data from different domains cannot be ignored. Our proposed model can solve the problem through the alignment of feature distribution between source and target domains and promote the performance in a new target domain.

7.6. Effect of ε

The effect of hyperparameter ε was also investigated, $\varepsilon = \{1, 2, 4, 8, 16\}$ were tested separately, and the obtained average value of *k* in the cross-channel, cross-subject and channel and cross-dataset experiments after using the proposed transfer method were calculated to evaluate the effect. The gradient caused by the back propagation of

the domain adaptation network will be passed to the feature extractor under the control of λ, of which the adjustment by ε is illustrated in Fig. 4 (a). A dynamic adjustment strategy was adopted. In other words, a smaller λ was generated in the initial stage, and as the number of iterations increased, λ was increased and approached at 1.0, which means that the passed gradient was completely received by the feature extractor to update its parameters.

Fig. 4 (b) shows the effect of ε on the average k of the three group experiments. A ε of<2 or greater than 8 has a negative impact on performance. Therefore, the ε was finally set to 4. One explanation is that a relatively small ε can ignore the domain adaptation constraint [22] and thus reduce the impact of the domain alignment, resulting in the feature extractor being unable to generate an appropriate representation with sufficient generalization to the target domain. In addition, the feature extractor is still unstable in the initial stage; thus, an overly large ε can make it easy for the feature extractor to neglect the information related to the classification task and focus more on domain adaptation, which may render the classifier difficult to converge.

7.7. How the proposed model works

The effectiveness of the proposed model is reflected in two elements: (1) the model can play a role of regularization and can effectively prevent over-fitting; and (2) the model can further improve model performance by reducing distribution discrepancies between domains.

7.7.1. Regularization effect

We adjusted the generalization performance of the model by changing the dropout rate, and we calculated k on the training set and test set before and after transfer learning to verify how the model works. It is exceptionally time-consuming to validate the model for all scenarios, so we only selected four different experimental scenarios to verify the effectiveness of the proposed transfer model. We chose the first item in Table 3, (i.e., EDF-set Fpz-Cz → EDF-set Pz-Oz), the last item in Table 4 (i.e., SH-set C3-A2 → SH-set C4-A1) and the first and last items in Table 5, (i.e., EDF-set Fpz-Cz → SH-set C4-A1 and SH-set C3-A2 → EDF-set Pz-Oz). The dropout rate of the feature extractor was increased from 0.1 to 0.5 in the step of 0.1, and then the k of the model on both of the training and test sets were calculated. As illustrated in Fig. 5, when the dropout rate is the same, the k before transfer on the training set is higher than that of the transfer model, while on the test set, it is generally lower than that of the transfer model. With an increase in the dropout rate, k on the training set decreased significantly, but that on the test dataset increased, and the model performance difference between the training dataset and test dataset tends to become smaller, indicating that the risk of over-fitting is reduced.

In addition, as the dropout rate increases, the decrease rate of k in the transfer model on the training set becomes smaller than that of the non-transfer model, and the k in the test dataset is significantly higher than in the non-transfer model, indicating that the transfer model effectively reduces the risk of model over-fitting and improves model performance even with a small dropout rate. One explanation is that once the model has over-fitting, the features of the hidden layer will be more specific to the source training dataset; thus, the domain adaptation network will make it easy to recognize the difference in the input data from alien domains, thus forcing the feature extractor to generate features that make it easier to generalize between source and target domains in the subsequent optimization process.

7.7.2. Reduction of distribution discrepancy

When the dropout rate is 0.5, the performance of the model on the test set does not continue to improve, and the model reaches an optimal generalization. However, it should be emphasized that the transfer model still achieves better performance than the non-transfer model when the dropout rate is set to 0.5. To explore its essential cause, t-SNE [67] was applied to visualize the hidden layer feature distribution; an example in the cross-channel experiment is shown in Fig. 6. The feature distributions of the source and target domains have intuitive differences before using transfer learning. Meanwhile, the distribution in each category achieved better alignment via our proposed model, indicating that performance improvement is not only due to the reduction of the over-fitting risk benefiting from the domain adaptation network, but also from the distribution alignment of the source and target domain.

7.8. Ablation study

To verify the proposed method, particularly the specific contribution of $\mathscr{L}_{\text{joint}}$ and $\mathscr{L}_{\text{trans}}$ to the model, we conducted an ablation study in various scenarios. In the above different scenarios, to reduce the computational load, a total of eight scenarios were selected for comparison. In each scenario, the model performance results after adding different loss items were compared. Results are shown in Table 7.

If only the sleep stage classification loss $\mathscr{L}_{\text{cls}}$ was considered, and the domain adaptation network was excluded (it is also the non-transfer

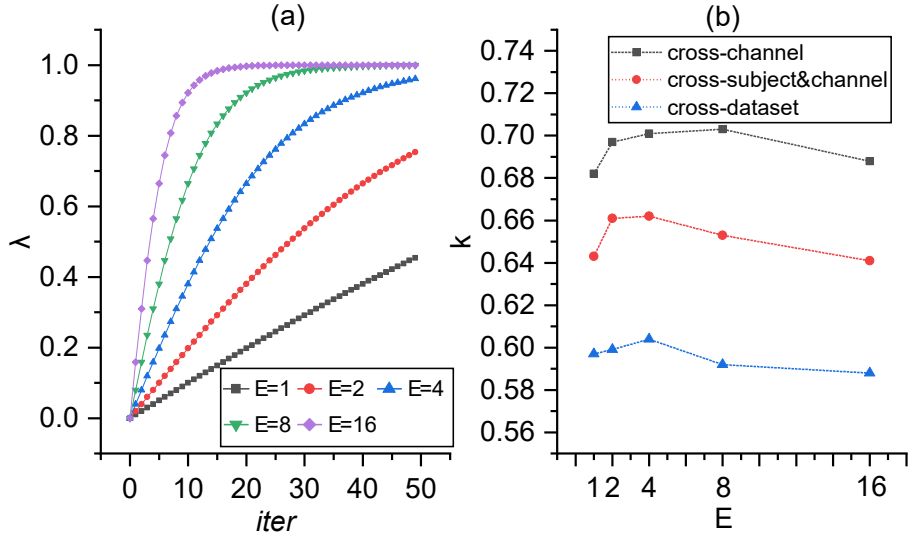


Fig. 4. Effect of ε to the model performance: (a) the λ adjusted by$\varepsilon = \{1, 2, 4, 8, 16\}$; and (b) average value of k in the cross-channel, cross-subject and channel, and cross-dataset experiments after using proposed transfer method with differentε.

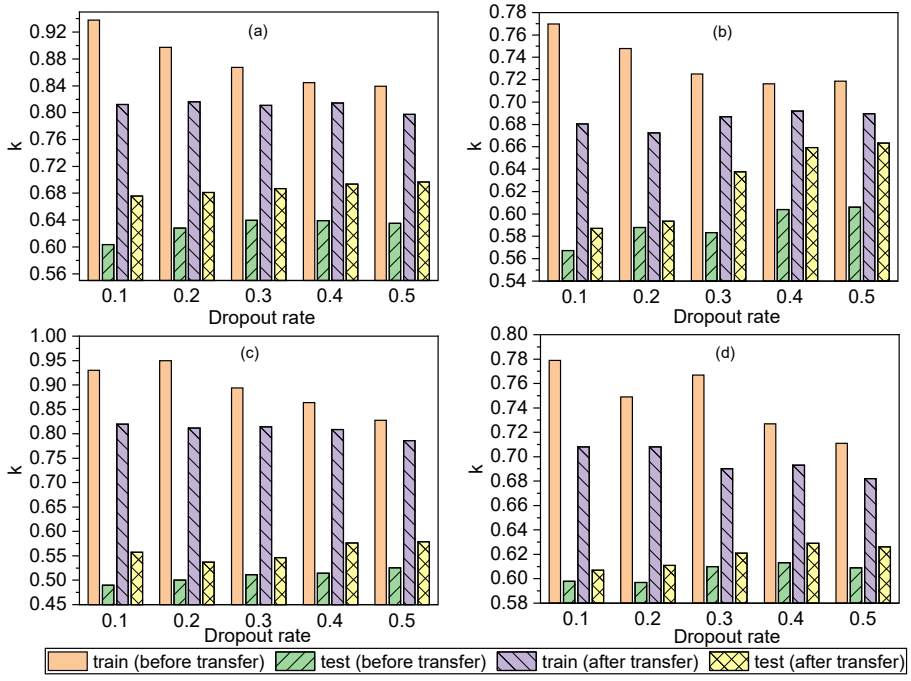


Fig. 5. Cohen's kappa coefficient (k) on the training dataset and test dataset before and after using transfer learning in four different scenarios with the dropout rate ranging from 0.1 to 0.5, including: (a) EDF-set Fpz-Cz → EDF-set Pz-Oz; (b) SH-set C3-A2 → SH-set C4-A1; (c) EDF-set Fpz-Cz → SH-set C4-A1; and (d) SH-set C3-A2 → EDF-set Pz-Oz.

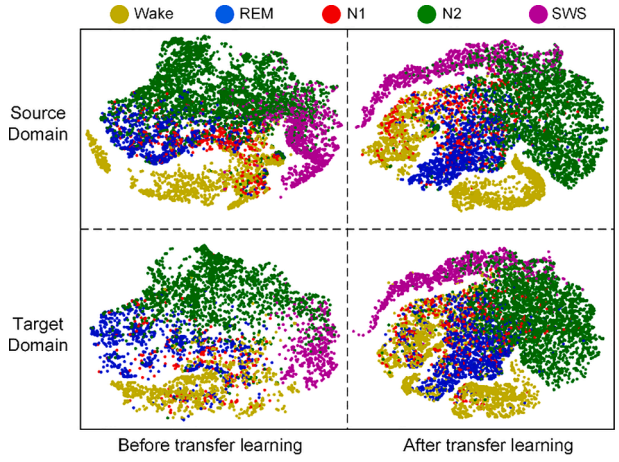


Fig. 6. Example of hidden layer feature distribution visualization using t-SNE framework in cross-channel experiment (EDF-set Pz-Oz → EDF-set Fpz-Cz).

model), the model performance was the worst. When $\mathscr{L}_{\text{joint}}$ was incorporated into the network, the performance achieved an average *Acc* improvement of 2.01 %, an average *MF*1 improvement of 1.70 %, and an average k improvement of 0.0313. In addition, when $\mathscr{L}_{\text{trans}}$ was added, the performance of the model can further achieve an average *Acc* improvement of 1.79 %, an average *MF*1 improvement of 1.20 %, and an average k improvement of 0.0203. In summary, both $\mathscr{L}_{\text{joint}}$ and $\mathscr{L}_{\text{trans}}$ can make a significant positive contribution to promoting performance.

Fig. 7 presents an example comparison of hypnograms on one subject in a cross-channel experiment (EDF-set Pz-Oz → EDF-set Fpz-Cz) and illustrates how the proposed method succeeded at promoting the performance of the task. It is evident in the dotted box that a non-transfer model using only $\mathscr{L}_{\text{cls}}$ yields a higher classification error, especially with numerous REM stages being misclassified to N1 or N2 stages. A transfer model using $\mathscr{L}_{\text{cls}} + \mathscr{L}_{\text{joint}}$increases the accuracy of the model significantly, and for the REM stage, they are less likely to be misclassified, especially at the N2 stage. Finally, the transfer model using $\mathscr{L}_{\text{cls}}+\mathscr{L}_{\text{joint}} + \mathscr{L}_{\text{trans}}$can further improve classification accuracy and the probability of REM being misclassified to N1 is further reduced.

However, there remain many stages that were misclassified to N1 or Wake stages. In previous sleep staging research, it was discovered that

Table 7
Ablation study result in a total of eight scenarios.

Source channel →target channel	$\mathscr{L}_{\text{cls}}$			$\mathscr{L}_{\text{cls}}+\mathscr{L}_{\text{joint}}$			$\mathscr{L}_{\text{cls}}+\mathscr{L}_{\text{joint}}+\mathscr{L}_{\text{trans}}$		
	Acc(%)	*MF*1(%)	k	*Acc*(%)	*MF*1(%)	k	*Acc*(%)	*MF*1(%)	k
EDF-set Pz-Oz → EDF-set Fpz-Cz[#]	76.6	66.3	0.679	77.8	67.9	0.700	79.1	68.1	0.713
SH-set C3-A2 → SH-set C4-A1[#]	76.8	61.0	0.665	79.3	64.6	0.707	80.1	64.5	0.720
EDF-set Pz-Oz → EDF-set Fpz-Cz[*]	74.0	63.1	0.638	76.0	64.5	0.673	77.8	65.0	0.691
SH-set C3-A2 → SH-set C4-A1[*]	72.2	55.9	0.606	74.1	60.4	0.639	76.6	60.5	0.663
EDF-set Pz-Oz → SH-set C4-A1	66.8	54.6	0.541	69.1	56.4	0.581	72.9	59.7	0.624
SH-set C4-A1 → EDF-set Pz-Oz	71.0	57.7	0.594	71.8	56.1	0.602	72.4	58.0	0.616
EDF-set Fpz-Cz → SH-set C3-A2	64.0	53.9	0.514	68.3	56.3	0.570	70.1	58.6	0.589
SH-set C3-A2 → EDF-set Fpz-Cz	71.4	58.8	0.607	72.5	58.7	0.622	74.2	60.1	0.640

[#] scenario in cross-channel experiment.
[*] scenario in cross-subject and channel experiment.

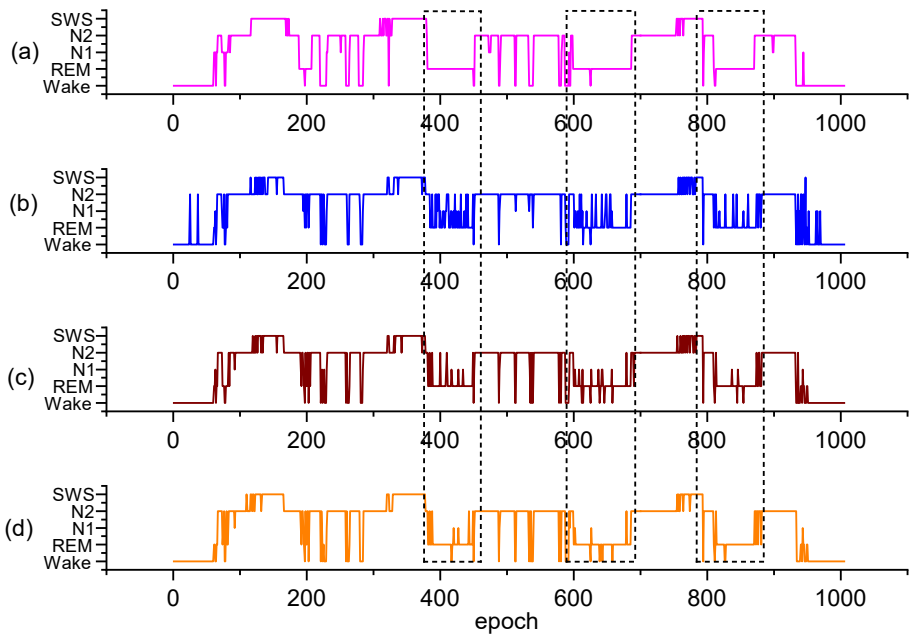


Fig. 7. Comparison of hypnograms on one subject in cross-channel experiment (EDF-set Pz-Oz → EDF-set Fpz-Cz). (a) Hypnogram labeled by human expert; (b) Hypnogram predicted by no-transfer model using only$\mathscr{L}_{\text{cls}}$; (c) Hypnogram predicted by transfer model using $\mathscr{L}_{\text{cls}}+\mathscr{L}_{\text{joint}}$; (d) Hypnogram predicted by transfer model using $\mathscr{L}_{\text{cls}}+\mathscr{L}_{\text{joint}}+\mathscr{L}_{\text{trans}}$.

the pairs REM-N1, REM-N2, and REM-Wake were relatively harder to distinguish. A reasonable explanation is that REM contains a clear eye movement pattern, but it is insufficient to expose the hidden pattern via only single-channel EEG without an EOG signal [4].

7.9. Comparison with DANN

Fig. 8 presents the comparison experiment with DANN [23]. S1 ∼ S8 are 8 scenarios consistent with the settings in the "Ablation study" section. The DANN mentioned in this study is obtained by deleting the GCL output in Fig. 1, so that the input of $D_J(\cdot)$ only contains hidden layer features without label prediction information. From the comparison between$\mathscr{L}_{\text{cls}}+\mathscr{L}_{\text{dann}}$and $\mathscr{L}_{\text{cls}}+\mathscr{L}_{\text{joint}}$, the performance of the six scenarios improved except for S1 and S4, implicating that the $\mathscr{L}_{\text{joint}}$ we proposed played a better role than DANN. And the comparison between $\mathscr{L}_{\text{cls}}+\mathscr{L}_{\text{dann}}+\mathscr{L}_{\text{trans}}$and $\mathscr{L}_{\text{cls}}+\mathscr{L}_{\text{joint}}+\mathscr{L}_{\text{trans}}$may further support this trend. The main reason is that the $\mathscr{L}_{\text{joint}}$ can transfer and integrate category-related prediction probabilities into a feature space, but DANN only considers the marginal distribution alignment of the source and the target domains, while ignores discriminative structures among

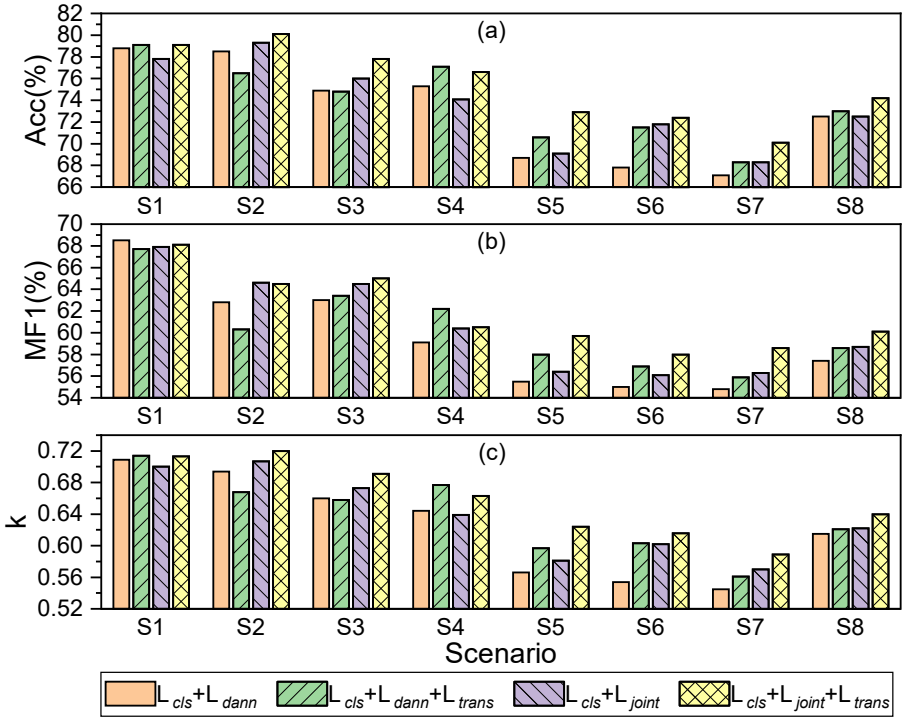


Fig. 8. The performance comparison between the proposed method and the DANN, the losses for comparison include $\mathscr{L}_{\text{cls}}+\mathscr{L}_{\text{dann}}$, $\mathscr{L}_{\text{cls}}+\mathscr{L}_{\text{dann}}+\mathscr{L}_{\text{trans}}$, $\mathscr{L}_{\text{cls}}+\mathscr{L}_{\text{joint}}$ and $\mathscr{L}_{\text{cls}}+\mathscr{L}_{\text{joint}}+\mathscr{L}_{\text{trans}}$, the performance indicators include (a) Accuracy (*Acc*); (b) Macro-averaging F1-score (*MF*1); (c) Cohen's kappa coefficient (k). S1: EDF-set Pz-Oz → EDF-set Fpz-Cz, S2: SH-set C3-A2 → SH-set C4-A1, S3: EDF-set Pz-Oz → EDF-set Fpz-Cz, S4: SH-set C3-A2 → SH-set C4-A1, S5: EDF-set Pz-Oz → SH-set C4-A1, S6: SH-set C4-A1 → EDF-set Pz-Oz, S7: EDF-set Fpz-Cz → SH-set C3-A2, S8: SH-set C3-A2 → EDF-set Fpz-Cz.

categories. Furthermore, from the comparison between $\mathscr{L}_{cls} + \mathscr{L}_{dann}$ and $\mathscr{L}_{cls}+\mathscr{L}_{dann}+\mathscr{L}_{trans}$, the performance of the six scenarios improved except for S2 and S3, indicating that $\mathscr{L}_{trans}$ can also exert a positive impact on performance in the hybrid structure of $\mathscr{L}_{trans}$ and DANN. The characteristics of the transition probability of sleeping adjacent epochs are fully considered by introducing $\mathscr{L}_{trans}$ in this study. This was implemented in the form of a loss function and well-integrated into the domain adaptation network, such that it could be synchronously optimized with feature extractors and classifiers in the optimization process. It can force the target domain and the source domain to have an optimally consistent transition probability distribution, and ultimately realize the constraint on the joint distribution of adjacent epochs of the target domain. This characteristic is unique in sleep staging tasks and almost non-existent in other similar classification tasks, such as image transfer tasks.

8. Conclusion

In the present study, we proposed a novel deep learning network to solve the problem of using labeled source datasets to assist in sleep staging in a new unlabeled target domain. The network consists of a feature extractor, a sleep stage classifier, and a domain adaptation network for cross-scenario automatic sleep stage classification based on single-channel EEG. Moreover, joint distribution loss and stage transition loss were proposed to achieve knowledge transfer between the source and target domain. Compared to similar existing methods, our method was implemented in an unsupervised manner and was more suitable for practical application scenarios. To clarify the effect of the model, we conducted cross-channel, cross-subject and channel, and cross-dataset experiments on two publicly available datasets. The experimental results demonstrated that our proposed model can play a role in regularization and can reduce distribution discrepancies between domains to enhance model performance. The ablation study indicated that both the joint distribution loss and stage transition loss can be effective, and they can play an even greater role in improving performance compared to the classic DANN method.

The main limitation of this study is that it was only verified on a single-channel EEG signal and did not include multi-channel information. In the future, we will include more EEG, EOG, and EMG information and conduct more verification with real datasets collected under diverse real-world scenarios.

CRediT authorship contribution statement

Zhengling He: Formal analysis, Software. **Minfang Tang:** Conceptualization. **Peng Wang:** Conceptualization, Validation. **Lidong Du:** Writing – review & editing, Methodology. **Xianxiang Chen:** Supervision, Writing – review & editing. **Gang Cheng:** Supervision, Funding acquisition. **Zhen Fang:** Project administration, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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